

Pharmacy – Medicine Sale and Billing System

Problem Statement

Develop a simple Pharmacy Medicine Sale and Billing System based on the following feature set: Medicine Details, Customer Details, Prescription Details, Medicine Availability, and Record Status.

1. Requirements

Write the functional requirements of the system based on the following features:

- Medicine Details: Medicine ID, Medicine Name, Category, Price, Quantity.
- Customer Details: Customer ID, Customer Name, Phone Number, Address.
- Prescription Details: Prescription Number, Doctor Name, Prescription Date, Prescribed Medicine.
- Check whether medicine is available or not.
- Update medicine quantity after sale.
- Record the status of the medicine/customer transaction.

2. Algorithm

Use simple numbered steps including:

Start → Enter Medicine Details → Enter Customer Details → Enter Prescription Details → Check Medicine Availability → If available continue sale → If not available display “Medicine Not Available” → Verify Prescription Details (if required) → Calculate Bill → Update Medicine Quantity → Store Sale/Record Status → Display Bill and Status → Stop.

3. Flowchart

Create a flowchart using standard symbols:

- Oval – Start/Stop
- Parallelogram – Input/Output
- Rectangle – Process
- Diamond – Decision
- Arrow – Flow Direction

The flowchart must show the decision “Is Medicine Available?” with Yes and No paths, prescription verification, bill calculation, quantity update, and record status.

4. ER Diagram

Create an ER Diagram with entities MEDICINE, CUSTOMER, PRESCRIPTION, SALE/BILL, and RECORD_STATUS. Show primary keys, foreign keys, important attributes, relationships, and cardinality (1:M, M:1, or 1:1 where appropriate). Keep the diagram simple and suitable for a 1st-year BCA student.

Suggested ER Structure

MEDICINE (Medicine_ID PK, Medicine_Name, Category, Price, Quantity)

1:M → SALE/BILL (Bill_ID PK, Medicine_ID FK, Customer_ID FK, Quantity_Sold, Total_Amount, Sale_Date)

M:1 → CUSTOMER (Customer_ID PK, Customer_Name, Phone, Address)

1:M → PRESCRIPTION (Prescription_ID PK, Customer_ID FK, Doctor_Name, Prescription_Date, Medicine_Name)